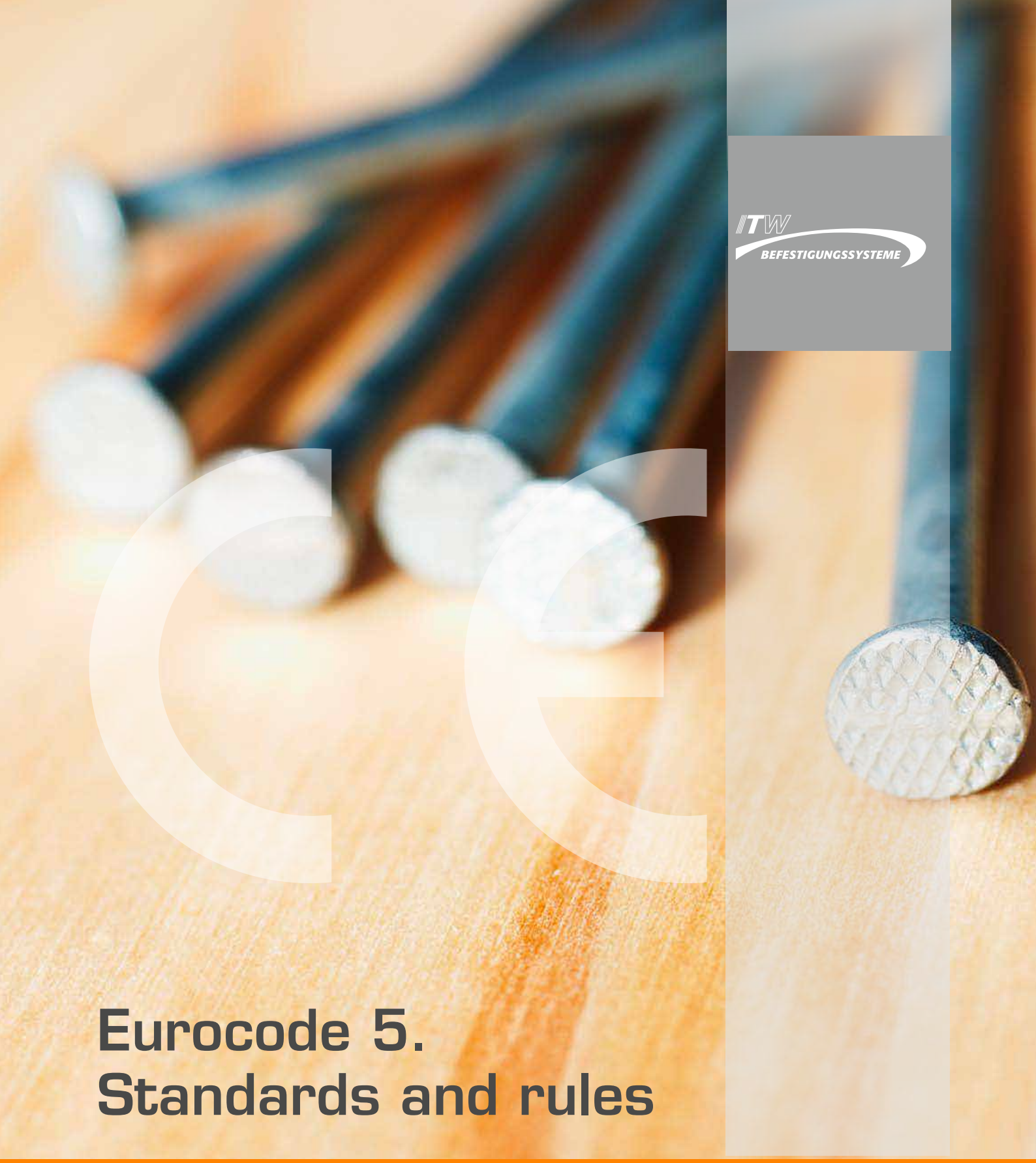

Eurocode 5. Standards and rules

General principles regarding fasteners

TW BEFESTIGUNGSSYSTEME spit haubold Paslode TOOLMATIC

The main purpose of this brochure is to provide professional groups such as civil engineers and architects with an increased need for information with a detailed overview of the standards and rules referred to in Eurocode 5 and the related standards.

ITW Befestigungssysteme GmbH belongs to the international group Illinois Tool Works Inc. (ITW) located in Glenview, Illinois. In 2012 the company celebrated its 100th anniversary. The company is among the top 100 patent owners in the US.

ITW operates in seven areas:

- Construction products
- Automotive industry
- Measuring technology/electrical engineering
- Welding equipment
- Gastronomic equipment
- Polymers and liquids
- Other business areas

ITW has 50,000 employees in 57 countries. The German subsidiary ITW Befestigungssysteme GmbH is located in Hemmingen, near Hanover. The main business area of ITW Befestigungssysteme GmbH in Germany is the sale and marketing of high-quality fasteners and fixing devices:

- for wooden structures, through the Paslode and haubold trademarks (nail and staple guns of pneumatic and impulse type);
- for the construction industry, through the Spit trademark (assembly equipment, diamond drilling, drilling and chiselling tools and dowel equipment);
- and for timber structures through the TOOLMATIC trademark (automation technology and process optimisation tools).

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I. General principles

1.1 Background

In the field of building construction, national standards have long restricted the free trade with building products. As early as 1975, the European Commission decided on an action program to remove trade barriers in the construction sector. The first Eurocodes for structural engineering were created in the 1980s. In 1989 the Commission delegated this task to the European Committee for Standardization (CEN).

The provisional European standards (ENV) were published and implemented on a trial basis as so-called National Application Documents (NAD). From 1997 these provisional standards were then converted into European standards (EN).

At the end of 2007, all 10 Eurocodes (design standards for the respective construction method) were published. Since 1 July 2012 these have been introduced in Germany building law and almost all national DIN standards have been withdrawn.

The objectives of the Eurocodes

- Uniform design criteria across Europe
- Uniform basis for research and development activities
- Harmonisation of different national regulations
- Simplification of tendering for construction work across Europe

Ten design standards

The programme of Eurocodes includes the following 10 standards, which usually consist of several parts:

EN 1990	Eurocode 0: Basis of structural design
EN 1991	Eurocode 1: Actions on structures
EN 1992	Eurocode 2: Design of concrete structures
EN 1993	Eurocode 3: Design of steel structures
EN 1994	Eurocode 4: Design of composite steel and concrete structures
EN 1995	**Eurocode 5: Design of timber structures**
EN 1996	Eurocode 6: Design of masonry structures
EN 1997	Eurocode 7: Geotechnical design
EN 1998	Eurocode 8: Design of structures for earthquake resistance
EN 1999	Eurocode 9: Design of aluminium structures

In addition to Germany, Eurocodes have been introduced by the standardisation organisations of the following countries: Austria, Belgium, Bulgaria, Estonia, Spain, Croatia, Ireland, Iceland, Italy, Greece, Cyprus, Lithuania, Luxembourg, Latvia, the Netherlands, Malta, Norway, Poland, Portugal, France, Sweden, Romania, Slovakia, Slovenia, Finland, Denmark, the Czech Republic, Hungary and the United Kingdom.

1.2 Areas of application of DIN EN 1995/Eurocode 5

Eurocode 5, equivalent to DIN EN 1995, applies to the design and construction of buildings and engineering structures made of wood (including solid wood or sawn, planed or round wood as well as glued laminated timber or wood-based structural products such as laminated veneer lumber structures) or to the design of structures from wood-based boards joined with adhesive or mechanical fasteners. It complies with the principles and requirements of EN 1990 on the safety and usability of buildings, as well as design and verification procedures.

Eurocode 5 consists of three parts:

1-1: General. General rules and rules for buildings

1-2: General rules. Structural fire design

2: Bridges

This document only deals with the general rules and regulations for the design of buildings (DIN EN 1995 1-1).

1.3 National annexes to DIN EN 1995-1-1

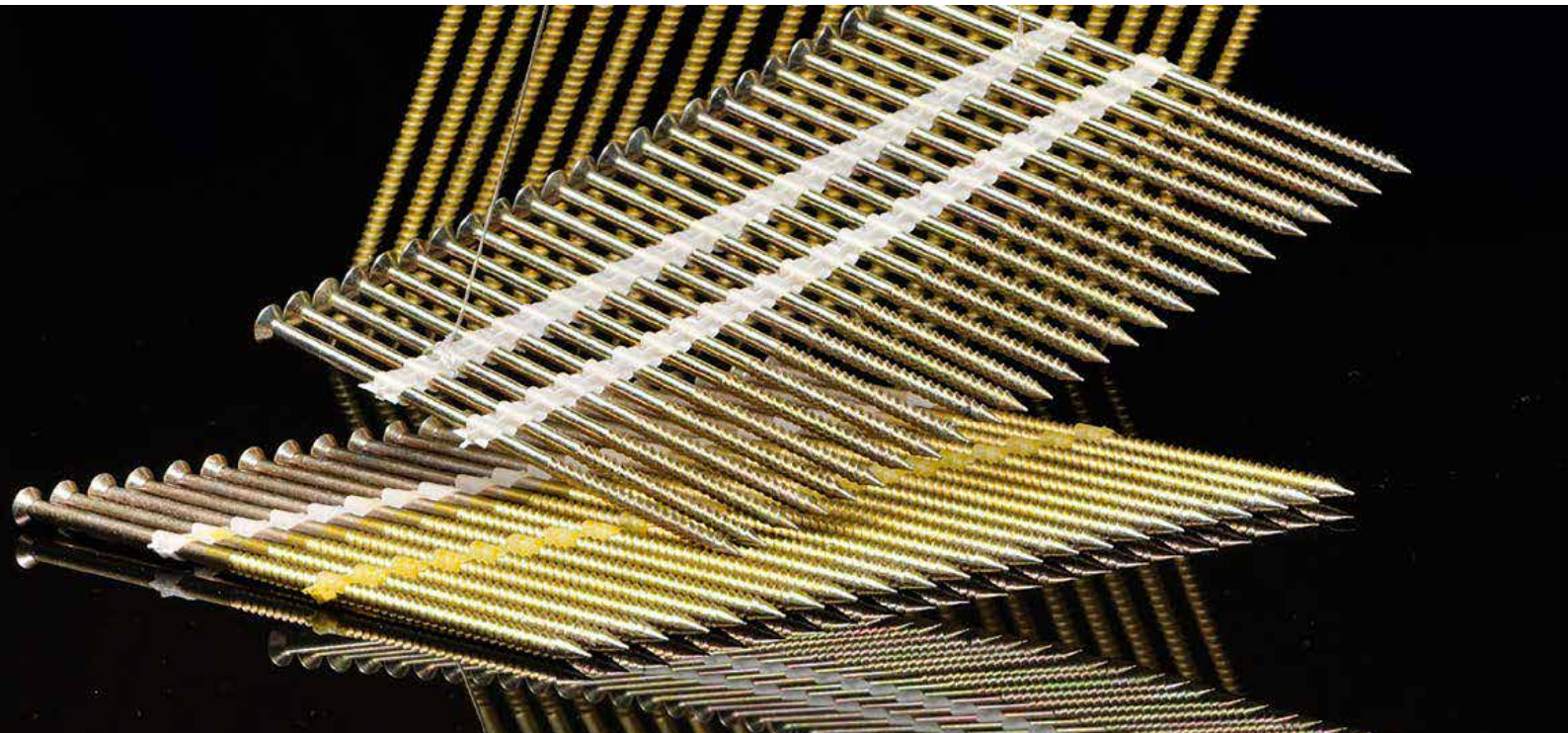
DIN EN 1995-1-1 includes alternative procedures, values and guidelines that can be selected at the national level. This standard is therefore only applicable with the corresponding so-called National Annex, which must be published by each member state. National parameters such as partial safety factors or climatic conditions are specified here. Each EU state has to decide on the content of its National Annex (which in Germany is called DIN EN 1995-1-1/NA).

National decisions are permitted in DIN EN 1995-1-1 for the following parts for example:

- determining load duration classes;
- determining service classes of load-bearing structures;
- partial safety factors of construction materials;
- wood-wood nail connections: instructions regarding nails in end grain;
- wood-wood nail connections: wood types that are sensitive to splitting;
- verification procedures for fasteners used to fasten to the wall.

In addition, each member state has the option of adding so-called NCI (non-contradictory complementary information) to the Annex.

In order to fully cover the area of application, in Germany **DIN 1052-10 i**s also used. It contains further national specifications on material properties as well as on the execution of and monitoring adhesive joints of wooden load-bearing structures.



1.4 Calculation principle according to DIN EN 1995-1-1

The national standardisation organisations have agreed that the calculations must be based on the limit state principle as a uniform concept, while the probability of failure is taken into account when determining the safety level.

In principle, two essential checks should always be carried out:

- checking the limit states of the load-bearing capacity – the structure must resist static maximum load;
- checking the limit states of usability – the user should not experience any discomfort during daily load and routine activities in the building, such as vibration or deformation of ceiling beams.

The building must also be safe and functional for people.

Calculation principle

The load-bearing capacity (R_d) of the construction elements used must always be greater than or equal to the effect of load (E_d).

$$E_d \leq R_d$$

The effects of load having a static impact on the structure are typically as follows:

- self-weight of the building materials;
- payloads of floor slabs (people, equipment);
- snow;
- wind.

The decisive effects of load are presented in the relevant parts of DIN EN 1991.

Natural deviations in construction timber

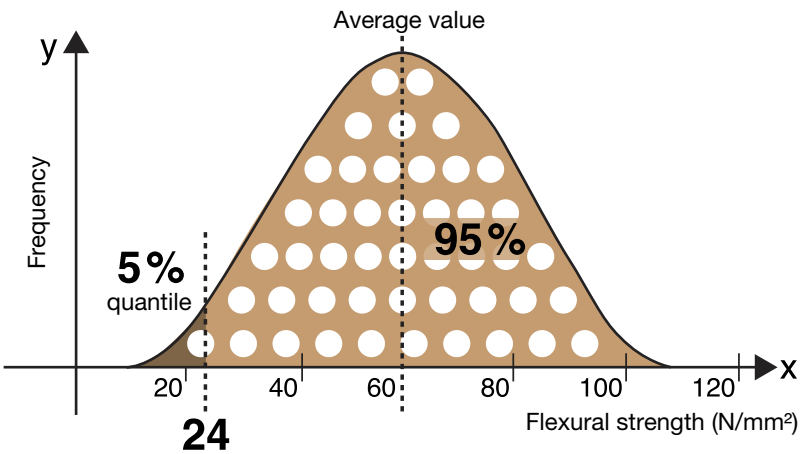
Strength and stiffness parameters depend on the wood's internal properties (e.g. density, knots, annular ring width, etc.) and on physical parameters (e.g. wood moisture content and load duration). The material parameters not only vary greatly between wood from different trees of the same species, but they vary even within the height of the trunk.

To take this variation into account, we base our calculations on the proofs on statistical distributions stated in DIN EN 1995-1-1, i.e. we use characteristic values for calculations. These so-called 5% fractile values are determined by tests (see an example below). Based on the statistical distribution, they correspond to the value, the smallest of which is present only in 5% of all examined cases. As a result of this strict requirement for building material, strength properties are decisive for load-bearing capacity and thus for the safety of people.

From characteristic value to calculated value

Since the strength and stiffness properties change significantly with changes in the moisture content of the wood or the load duration, the characteristic values determined in the laboratory must be reduced to the calculated values. To do this, it is necessary to assign a service class or load duration class to a specific calculated situation based on the condition of the building element.

Graphic: result of the examination of the flexural strength of softwood (material strength class C24) on 46 test specimen



The system of service classes

The service classes are determined based on the moisture content of the wood under certain environmental conditions. This is crucial for the calculations because the higher the wood moisture content, the lower the strength and stiffness properties of the wooden element or joint.

The service classes are defined as follows:

Service class 1
Moisture content of a building material corresponding to a temperature of 20°C and a relative humidity of the ambient air that exceeds 65% only for some weeks a year (for example, closed heated structures). This corresponds to an average moisture content of softwood ≤ 12%.

Symbol:
INDOOR ENVIRONMENT

Service class 2
Moisture content of a building material corresponding to a temperature of 20°C and a relative humidity of over 85% for only some weeks a year (for example, uncovered and covered structures and rear-ventilated areas in roof areas or masonry cladding).
This corresponds to an average moisture content of softwood ≤ 20%.

Symbol:
PROTECTED OUTDOOR ENVIRONMENT

Service class 3
Climatic conditions that lead to higher moisture content than in service class 2 (e.g. structures exposed to weather conditions). This corresponds to an average moisture content of softwood > 20%.

Symbol:
OUTDOOR ENVIRONMENT

Load duration classes

Wood and wood materials are characterised by high strength (stiffness) for short load durations, but under permanent load it is much less. For this reason, wooden parts and joints must be assigned to an appropriate class depending on their use.

The load duration classes are defined as follows:

	Load duration class	Accumulated duration of characteristic load	Load examples
1	Permanent	more than 10 years	self-weight
2	Long-term	6 months to 10 years	storage
3	Medium-term	1 week to 6 months	imposed floor load, snow
4	Short-term	less than one week	snow, wind
5	Instantaneous		wind and accidental

Modification coefficients k_{mod}

The correction value for the characteristic value, which is taken from a table in accordance with the service class and load duration class, is the so-called modification coefficient k_{mod}.

For example, for a plywood joint the modification coefficient value in case of service class 2 and short-term load is 0.90, which corresponds to a 10% reduction.

k_{mod} values for common wooden building materials

Building material	Service class	Load duration class: permanent	long-term	medium-term	short-term	Instantaneous
Solid wood, laminated wood, laminated veneer lumber	1	0.60	0.70	0.80	0.90	1.10
	2	0.60	0.70	0.80	0.90	1.10
Plywood	3	0.50	0.55	0.65	0.70	0.90
OSB/3, OSB/4	1	0.40	0.50	0.70	0.90	1.10
OSB/3, OSB/4	2	0.30	0.40	0.55	0.70	0.90
Chipboard P6, P7	1	0.40	0.50	0.70	0.90	1.10
Chipboard P7	2	0.30	0.40	0.55	0.70	0.90

For joints of wooden parts with different modification factors, the average is calculated as follows:

$$k_{mod} = \sqrt{k_{mod,1} \cdot k_{mod,2}}$$

If a load combination consists of loads that belong to different load duration classes, a value of k_{mod} that belongs to the load with the shortest duration should generally be used.

Partial safety factors γ_M

Due to inaccuracies in statistical studies, the Eurocode calculation concept provides for an additional reduction through the use of the so-called partial safety factor.

As a rule, the partial safety factor γ_M for wood, wood-based materials and fasteners is 1.3.

Calculation value

The parameter characterising strength, for example, the flexural strength calculated value R_d, is ultimately calculated as follows:

$$R_d = \frac{k_{mod} \cdot R_k}{\gamma_M}$$

Where

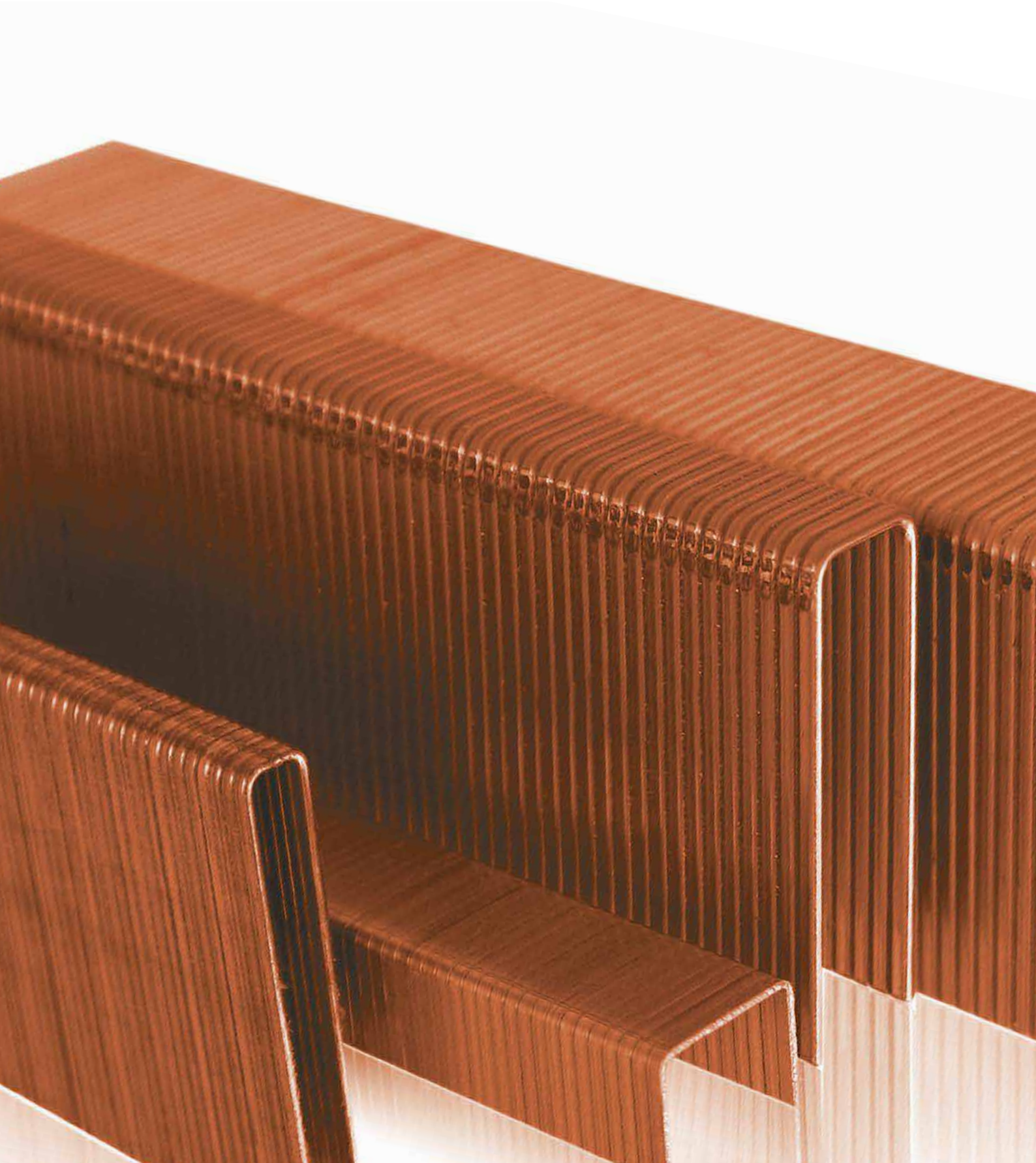
- R_k = characteristic value of a strength parameter (determined through tests)
- k_{mod} = the modification coefficient based on load duration class and moisture content
- γ_M = partial safety factor for one property of the building material

1.5 Ensuring the safety level according to DIN 1052 in Germany

When drawing up the European standards, the German Federal Ministry of Building and Building Supervision required that the level of safety be at least in accordance with DIN 1052.

For this reason, very detailed NCI information has been added to the national annex and, in particular, DIN 1052-10, and a number of application standards have been published.

II. Fasteners



2.1 Conformity and CE marking

Load-bearing structures usually consist of various elements/construction elements (e.g. ceiling beams, nails, dowels, beam shoes, etc.). For each individual product put on the market for this purpose, manufacturers must declare their performance using a specific product standard (e.g. DIN EN 14592) or European Technical Assessment.

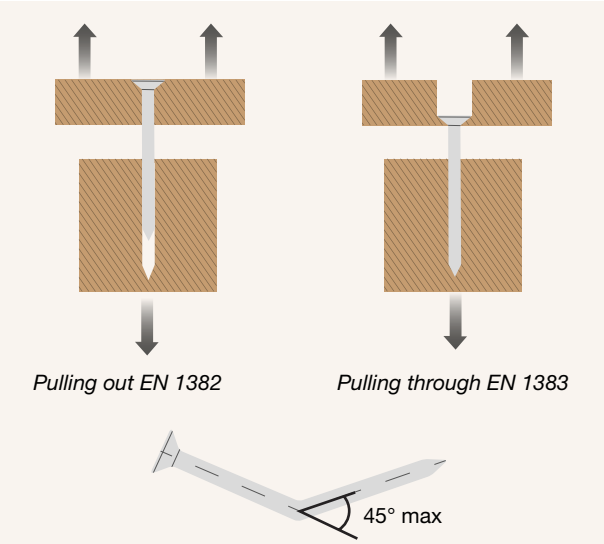
Various tests carried out by independent testing institutes may also be used for this purpose. This ensures that the product meets the essential requirements of the German Building Code (BauPVo).

After this, the manufacturer confirms with a Declaration of Performance (DoP) that their product complies with the current European regulation for the specific product. Such product may be marketed and sold on the European market provided that the CE marking is clearly visible on it. CE stands for “Conformité Européenne” (European conformity) and is a visual confirmation of conformity.

2.2 DIN EN 14592

Eurocode 5 prescribes that metal fasteners must comply with DIN EN 14592. The product standard specifies requirements and test methods as well as quality control. DIN EN 14592 only applies to steel fasteners and includes nails, staples, screws, belaying pins and bolts with nuts.

In accordance with the requirements of DIN EN 14592, it is necessary to carry out an Initial Type Testing, i.e. a test of mechanical properties, such as the behaviour of the head during pulling out and pulling through, tensile strength of the wire, bending moment) and to implement a quality management system.



Pulling out EN 1382 *Pulling through EN 1383*

Bending test: EN 409

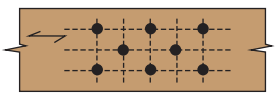
The exception in Germany is that the requirements for the use of dowel fasteners are specified in accordance with DIN EN 14592 in DIN 20000-6. Designers therefore have the option, among other things, of assigning a load-bearing class to profiled nails (formerly special nails).

2.3 Wood joints with dowel fasteners

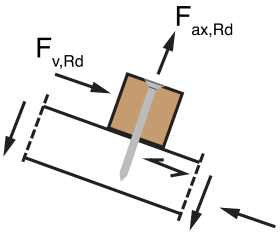
Eurocode 5 deals with wood joints with metal fasteners in Chapter 8. The rules that apply to them are explained below.

The location of the fasteners, their size and distances, as well as the distance from the edges and ends, must be chosen so as to ensure the prescribed load-bearing capacity and stiffness.

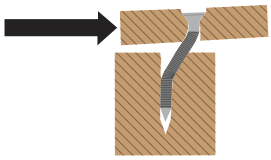
It should be noted that the load-bearing capacity of several joints of the same type and dimensions may be less than the sum of the load-bearing capacities of each individual fastener.



Distances from edges



Combined load

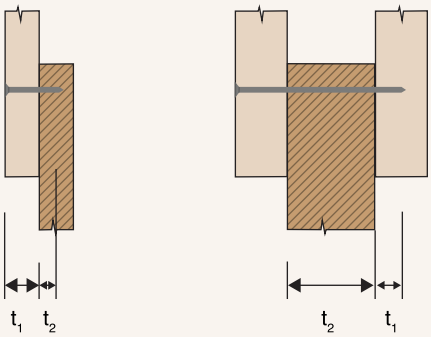


Shear

Typical applications of nails, staples and screws:

- open boarding on rafters
- rafters on purlins (only nails and screws)
- reinforcing cladding on a wooden frame
- steel on wood, e.g. assembly tape, angles, beam shoes (nails and screws only)
- wooden cladding for facades

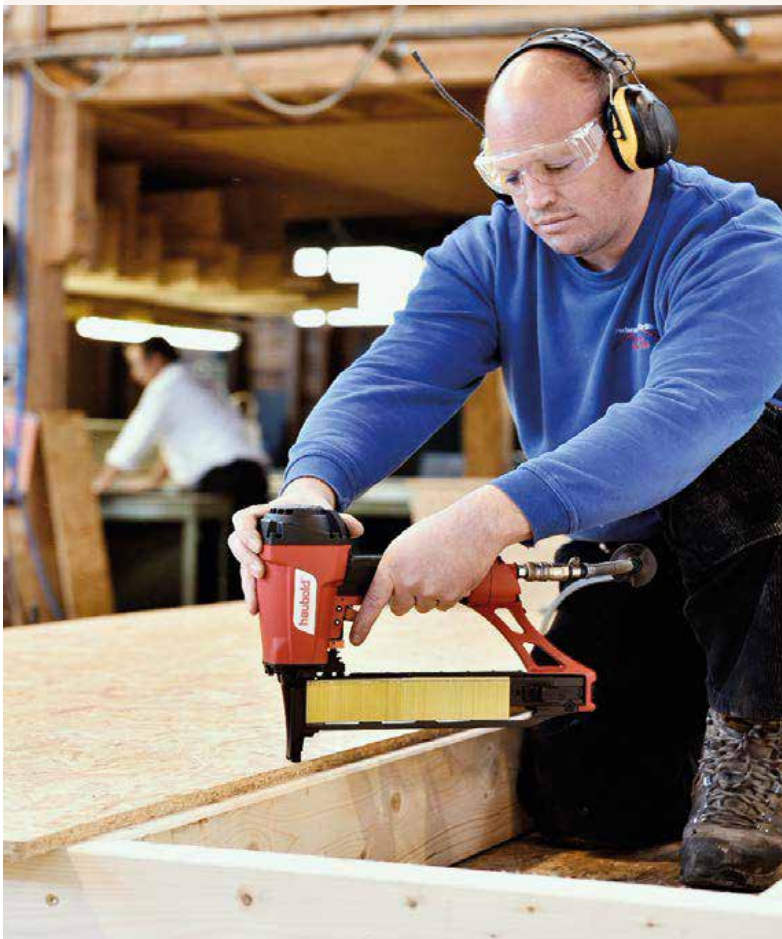
Determination of a single-cut (a) and double-cut (b) joint:



(a) (b)

t_1 = thickness of the top element (a)
thickness of the top element or penetration depth (b)
t_2 = penetration depth (a)
thickness of the middle element (b)

The joint must contain at least two nails.



Penetration depth of the nail head into wood or chipboard

Unless otherwise specified, nails/staples should be driven in at right angles and to such a depth that the head of the nail/staple back is flush with the surface.

When joining chipboards, the nail heads/staple backs must not be recessed over 2 mm. In this case, the minimum thickness of wood materials must be increased by 2 mm.

Penetration depth of the nail head into wood material elements

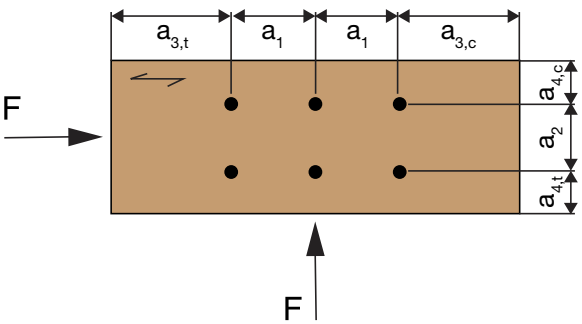
Wood-wood joints

Minimum distances between dowel fasteners

The calculated load-bearing capacity of wood joints can only be used if the corresponding minimum distances are observed.

In the case of dowel fasteners, these are determined as follows:

- a_1 distance along grain
- a_2 distance perpendicular to grain
- $a_{3,t}$ loaded end
- $a_{3,c}$ unloaded end
- $a_{4,t}$ loaded edge
- $a_{4,c}$ unloaded edge



Minimum distances and penetration depths for softwood

The simplified summary of the minimum distances between fasteners and the minimum penetration depths that depend on the diameter d are shown in the table.

The following assumptions were taken into account:

- wood density $\rho_k \leq 420$ kg/m³;
- no preliminary holes have been made for fasteners
- nail (d < 5 mm);
- screws (5 mm ≤ d ≤ 6 mm);
- staple angle in grain direction ≥ 30 °, measured to the middle of the staple back

Shear	Nails smooth	Nails profiled	Staples	Screws
Penetration depth in wood	8d	6d	14d	6d
Minimum distance to the loaded edge	7d	7d	20d	10d
Minimum distance to the unloaded edge	5d	5d	10d	5d
Minimum distance to the loaded end	15d	15d	10d	15d
Minimum distance to the unloaded end	10d	10d	15d	10d
Minimum distance along grain	10d	10d	15d	12d
Minimum distance in the direction in 90° angle with grain	5d	5d	15d	5d
Pull out				
Penetration depth in wood	12d*	8d	14d*	6d
Minimum distance to the unloaded edge	5d	5d	10d	4d**
Minimum distance to the unloaded end	10d	10d	15d	10d**
Minimum distance along grain	5d	5d	10d	7d**
Minimum distance in the direction in 90° angle with grain	5d	5d	15d	5d**

** Only for loads of medium, short or instant duration.*
*** If the thickness of the wood element is ≥ 12d*

Distances from edges for board materials

According to the German national annex for board materials, the following edge distances apply:

Distances from edges for board materials	unloaded edge	loaded edge
Plywood	3d	7d
OSB-boards		
Chipboards glued with synthetic resin		
Class HB.HLA2 chipboards		
Gypsum boards	7d	10d
Fibre gypsum boards (here Fermacell)	4d	7d

For staples, these distances are valid to the nearest leg.

Puitu tuleb ette puurida auk, kui:

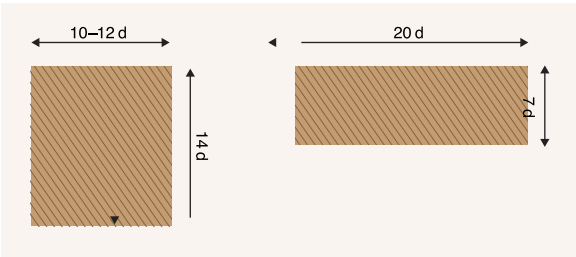
- the characteristic density of the wood is ≥ 500 kg/m³;
- nail diameter is > 6 mm;
- the thickness of the pine wood element is less than ...

$$t = \max\left\{\begin{matrix} 7d \\ (13d - 30)\dfrac{\rho_k}{400} \end{matrix}\right\}$$

- the thickness of an element from other wood species is less than ... .

$$t = \max\left\{\begin{matrix} 14d \\ (13d - 30)\dfrac{2\rho_k}{00} \end{matrix}\right\}$$

(In the case of edge distance $a_4 \geq 10d$ ($\rho_k \leq 420$ kg/m³) or for construction elements such as timber formwork, load-bearing or counter battens, the formula used for pine wood can be chosen.)



Drawings of minimum cross-sections of solid wood elements

2.4 Calculation of timber structures

Wood joints can be calculated for the following cases:

a) **shear**, determined by using various failure mechanisms;
b) **pull out**, pull out from the base element or pull through from a fastened element;
c) a **combination** of both cases.

a) Shear

The bearing capacity shear of a joint is characterised by a combination of compressive stress, flexural strength, and a parameter characterising pull out ("rope effect").

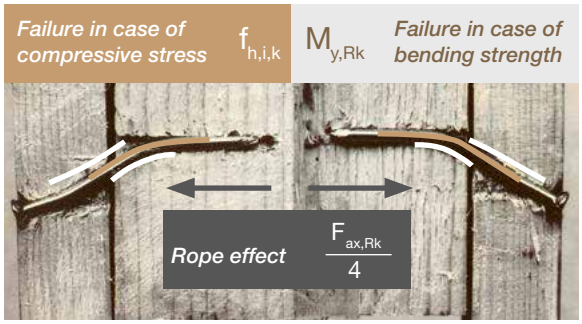
Compressive stress. This term refers to the strength of the wood resisting the pressure of the metal fasteners. Typically, the resistance to compressive stress changes when the force impact angle changes in relation to the wood grain. As wood is softer than steel, in the event of a failure, the wood around the fastener will fail and a cavity is formed.

Flexural strength. When wood elements move in relation to one another, the fastener may bend in the event of breakage.

Rope effect. In the event of a rupture, the bent fastener is pulled out of the wood like a rope, and a cavity remains.

The impact of the rope effect on the load-bearing capacity must be limited to the following values:

- round nails 15% (possible only for medium, short-term or instantaneous load duration class!);
- grooved nails 50%
- screws 100%.



Quelle: TU Delft, NL

a) Shear (continuation)

Six failure cases were calculated for single-cut joints, of which the failure case with the lowest value is decisive. In the case of thin fasteners, the formula for determining failure method (f) usually applies:

$$F_{v,Rk} = 1{,}15 \cdot \sqrt{\frac{2\beta}{1+\beta}} \cdot \sqrt{2 \cdot M_{y,Rk} \cdot f_{h,1,k} \cdot d} + \frac{F_{ax,Rk}}{4}$$

Where

$\mathbf{F_{v,Rk}}$ ⇒ characteristic load-bearing capacity for one shear surface and fastener [N]

β ⇒ relation of compressive stresses between construction elements

$\mathbf{d}$ ⇒ fastener diameter [mm]

$\mathbf{M_{y,Rk}}$ ⇒ characteristic value of the yield of the fastener [Nmm]

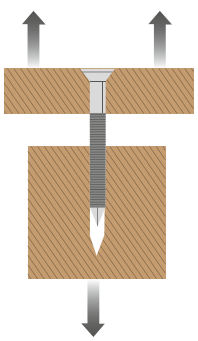
$\mathbf{f_{h,1,k}}$ ⇒ characteristic compressive stress in a wood element [N/mm²]

$\mathbf{F_{ax,Rk}}$ ⇒ characteristic pull out resistance of the fastener [N]

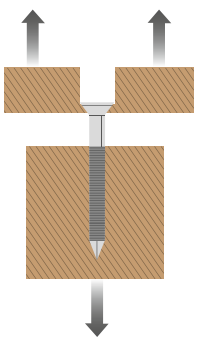
The ITW software reliably calculates the load-bearing capacities of shear and pull properties for wood-wood, wood material-wood, plaster-wood and sheet steel-wood joints in accordance with EN 1995-1-1.

b) Pull out

Pull-out fasteners are pulled out from the base elements or pulled through the fastened element. The process carried out first determines the load-bearing capacity of the entire joint.



Pull out



Head pull out

Typical loads in the case of pull out

- Short-term load (wind load): fixing the external facade/battens
- Long-term or permanent pull out: fastening the support batten of gypsum boards under ceilings or pitched roofs

Requirements for nails and staples for pulling out

Smooth nails should not be used for permanent and long-term loads. In the case of staples covered with polymer resin, it is permitted provided ETA exists. For profiled nails, only the length of the profiled part of the nail must be taken into account in the formula. Nails at the ends are not suitable for transmitting force in the direction of the nail.

Calculation of the characteristic pull-out resistance

The characteristic value of the-pull out resistance $F_{ax,Rk}$ should, as a rule, be the lowest value obtained from the following formulas and multiplied by the effective number of fasteners.

For profiled nails

$$F_{ax,Rk} = \min \begin{Bmatrix} f_{ax,k} \cdot d \cdot t_{pen} \\ f_{head,k} \cdot d_h^2 \end{Bmatrix}$$

For screws

$$F_{ax,\alpha,Rk} = \min \begin{Bmatrix} \dfrac{f_{ax,k} \cdot d \cdot l_{ef} \cdot k_d}{1{,}2\cos^2\alpha + \sin^2\alpha} \\ f_{head,k} \cdot d_h^2 \left(\dfrac{\rho_k}{\rho_a}\right)^{0,8} \end{Bmatrix}$$

For smooth nails/staples

$$F_{ax,Rk} = \min \begin{Bmatrix} f_{ax,k} \cdot d \cdot t_{pen} \\ f_{ax,k} \cdot d \cdot t + f_{head,k} \cdot d_h^2 \end{Bmatrix}$$

Where

$\mathbf{F_{ax,Rk}}$ ⇒ characteristic pull out resistance of the fastener [N]

$\mathbf{f_{ax,k}}$ ⇒ characteristic value of the pull-out strength [N/mm²]

$\mathbf{f_{head,k}}$ ⇒ characteristic value of the head pull through [N/mm²]

$\mathbf{t_{pen}}$ ⇒ penetration depth or the length of the profiled part [mm]

$\mathbf{d}$ ⇒ fastener diameter [mm]

$\mathbf{t}$ ⇒ thickness of the construction element closer to the nail head [mm]

$\mathbf{d_h}$ ⇒ diameter of the fastener head [mm]

Density correction

For the pull-out test, profiled nails can be assigned to one of the load-bearing classes according to DIN 20000-6. Thus the designer can recalculate the parameters $f_{ax,k}$ and $f_{head,k}$ according to the actual characteristic density ρ_k. Thereat ρk values in the case of wood materials are 380 kg/m³ and for solid wood 500 kg/m³.

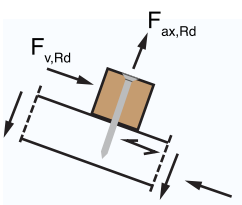
According to DIN EN 1995-1-1/NA, the tested staples can be considered as two load-bearing profiled nails of load-bearing class 2 for medium or short axial load duration if the staples are coated with a polymer resin layer.

Nail type	$f_{ax,k}$	Nail type	$f_{head,k}$
Load-bearing capacity class of profiled nails	–	Load-bearing capacity class of profiled nails	–
1	$30 \cdot 10^{-6} \cdot \rho_k^2$	A	$60 \cdot 10^{-6} \cdot \rho_k^2$
2	$40 \cdot 10^{-6} \cdot \rho_k^2$	B	$80 \cdot 10^{-6} \cdot \rho_k^2$
3	$50 \cdot 10^{-6} \cdot \rho_k^2$	C	$100 \cdot 10^{-6} \cdot \rho_k^2$
–	–	D	$120 \cdot 10^{-6} \cdot \rho_k^2$
–	–	E	$140 \cdot 10^{-6} \cdot \rho_k^2$
–	–	F	$160 \cdot 10^{-6} \cdot \rho_k^2$

Pull-out parameter $f_{ax,k}$ and head pull-through parameter $f_{head,k}$ characteristic values for nails (N/mm²)

c) Combined load

In the case of joints where the load is expressed as a combination of the axial ($F_{ax,Ed}$) and the component perpendicular to the axis ($F_{v,Ed}$) of the nail, the following condition must be met:



$$\left(\frac{F_{ax,Ed}}{F_{ax,Rd}}\right)^m + \left(\frac{F_{v,Ed}}{F_{v,Rd}}\right)^m \leq 1$$

m = 1 for smooth nails or staples
m = 2 for grooved nails or screws

This means that if wood is loaded in one direction (laterally or axially), it is not possible to carry the full load in another direction.

2.5 Corrosion protection

Metal fasteners must either be corrosion resistant by nature or be protected against corrosion to ensure a long service life.

DIN EN 1995-1-1 provides examples in the form of a table of minimum corrosion protection or requirements for building materials for different service classes:

Fastener	Service classes*		
Nails and screws, d ≤ 4 mm	not required	Fe/Zn 12c**	Fe/Zn 25c**
Bolts, belaying pins, nails and wood screws, d > 4 mm	not required	not required	Fe/Zn 25c**
Staples	Fe/Zn 12c**	Fe/Zn 12c**	stainless steel

** Thicker hot-dip galvanised or stainless-steel coatings must be taken into account, especially in corrosive conditions.*
*** For hot-dip galvanised coatings, Fe/Zn 12c must normally be replaced by Z275 and Fe/Zn 25c by Z350 in accordance with EN 10147.*

Wood preservatives and wood species must also be taken into account, as their ingredients may react with different coatings. For example, it is always recommended to use stainless-steel fasteners for larch or spruce.

In addition, certain environmental conditions (salty air in coastal areas, higher humidity in swimming pools) may require the use of thicker zinc layers or special stainless steel.

ITW colour codes

To make it easier to choose the right fastener, ITW has developed colour codes that indicate the service class of the fasteners and the type of coating.

Service class 1

Uncoated

Indoor environment

- uncoated steel in dry heated buildings of service class 1

Galvanised

Indoor environment

- various zinc layers up to 7 µm thick

Service class 2

Galvanised, 12 µm

Protected outdoor environment

- 12 µm zinc layer for service class 2 meets the requirements of Eurocode 5 for nails and staples

Galv-Plus™

Protected outdoor environment

- 15 µm thick zinc-aluminium alloy coating developed by ITW
- 70% better corrosion protection
- for service class 2, exceeds the requirements of Eurocode 5 for nails

Service class 3

Hot galvanised

Outdoor environment

- minimum layer thickness 55 µm
- for service class 3, meets all requirements for outdoor use

Stainless steel

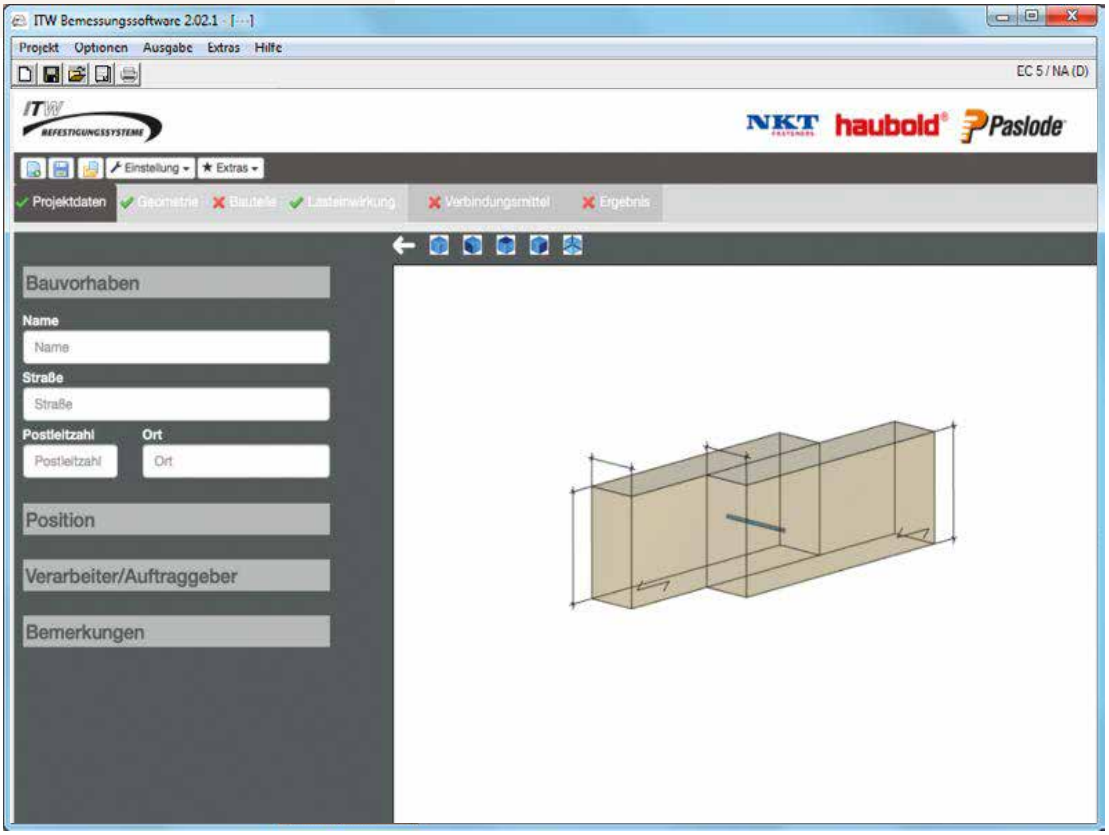
Outdoor environment

- stainless steel fasteners are manufactured in quality class A2 as well as A4
- long service life, especially in corrosive conditions

2.6 ITW calculation software

To simplify the use of complex rules and formulas, ITW offers calculation software that can be downloaded for free from our website. This software can be used to calculate the load-bearing capacity of dowel fasteners (wood-wood, wood material-wood, gypsum material-wood and sheet steel-wood joints) in accordance with DIN EN 1995-1-1. Calculations can be made for staples, nails and screws.

In addition, ITW calculation software provides tips for using the tools with important background information. The printout of the document also provides additional information on the load-bearing capacity of the fastener and an overview of the required distances from the edge.



III. Sources and additional information

Sources

- Das neue Sicherheitskonzept der DIN 1052 (Holzbau-Quadriga 1/2005)
- DIN EN 1995-1-1:2004+AC:2006+A1: 2008+A2:2014
- DIN EN 1995-1-1/NA:2013-08
- DIN EN 14592:2008+A1:2012
- DIN 20000-6:2013-08
- DIN 1052-10:2012-05
- DIN 1052:2008-12

Additional information

On our websites **www.itw-befestigungssysteme.de** and **www.itw-eurocode5.de** we further information can be found about Eurocode 5 and load-bearing capacity calculation software. Of course, our specialists as well as the author of this brochure will be happy to advise you.

Author:

Certified engineer (FH) Klaas Gümmer
Member of the DIN Standards Committee Holzbau (wooden structures)

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Sales in Germany I Austria
ITW Befestigungssysteme GmbH
Carl-Zeiss-Straße 19
D - 30966 Hemmingen
Phone: +49 (0) 51 142 040
Fax: +49 (0) 511 420 4206
info@itw-befestigungssysteme.de
www.itw-befestigungssysteme.de

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